MINING DEALER BEST PRACTICE

Crankshaft/ Camshaft Washing Rack for Turntable Washer

Maintenance and Repair | Application | Component Life Management | **Component Renewal (CRC)** | MARC Management

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July 2015
1506-4.03-1382

1.0 Introduction

Borusan Makina Kazakhstan has designed a washing rack to wash crankshafts and camshafts safely and efficiently in the turntable washer. The rack, as built, is applicable for all engine models up to 3508 length (with appropriately fabricated retainers and adapters).

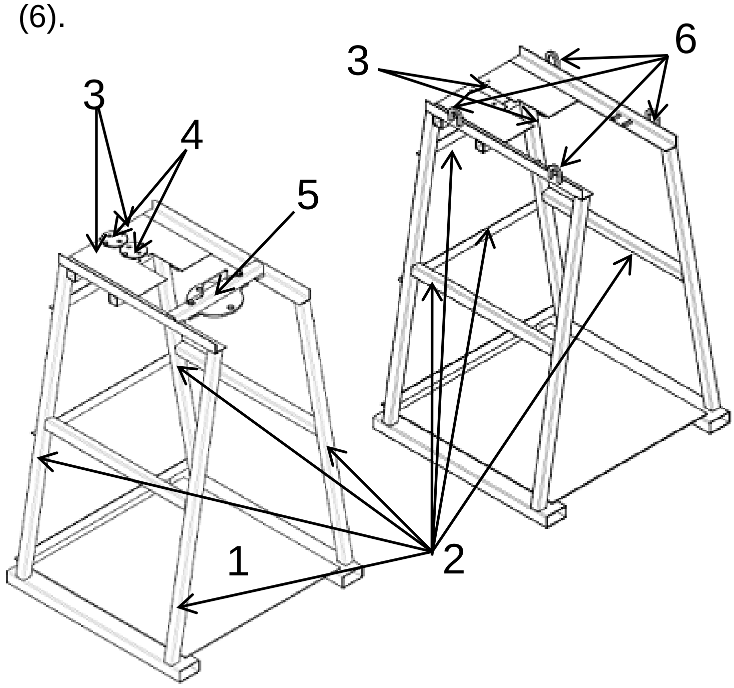
Previously it was tiresome to wash crankshafts and camshafts in the turntable washer. Crankshafts and camshafts were put onto pallets and fastened to them. These components might fall during washing in the washer and be damaged. Besides, they were not washed well.

According to the washing standard procedure, components should be washed in baskets or on stands/racks. Thus, this washing rack allows safe and efficient washing of the crankshafts and camshafts and following the procedure required.

2.0 Best Practice Description

The washing rack for crankshafts and camshafts is designed in a convenient way. It has the following main elements:

- Screen bottom to allow the better washing of the components(1),
- Reliable supports to provide the rigidity of the structure (2),
- 2 x Camshaft Plate to hang camshafts in the stand. There are two widths of the slot for camshafts. It was designed to use the wider slot to wash camshafts with gears installed (3);
- 2 x Camshaft Retainer Plate prevents the camshafts from banging into each other and sliding from the rack during the lifting (4);
- Crankshaft Adapter removable that securely keeps the crankshaft in the rack (5);
- 4 x Lifting eyes to lift the rack by the crane and put/remove it into/from the turntable washer (6).



Crankshaft & Camshaft Washing Rack

The washer work height is 1830mm (6 feet) and the turntable diameter is 1575mm (5.17 feet).

This rack enables you to wash several components simultaneously (1 x Crankshaft, 2 x Camshafts and Valve Mechanism) in the turntable washer that allows to utilize the washer efficiently.

3.0 Implementation Steps

<u>The Crankshaft and Camshaft Washing Rack is in use</u>.

It is simple to use this rack.

1. Hang 1 x Crankshaft and 2 x Camshafts in the rack.
2. Lift the rack by the crane, carry it to the washer and put it on the turntable.
3. Wash the components. The inhibitor is already in the aqueous solution of the washer.
4. Then take the rack out from the washer and leave the components to get cold.

Crankshaft & Camshaft Washing Rack in use

4.0 Benefits

- Components to be washed are safely fastened.
- Efficient utilization of the washer (several components) are washed at the same time.
- Components are washed better.

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5.0 Resources Required

- Machine Shop.

6.0 Supporting Attachments / References

The washing rack's detail and assembly drawings are attached.

7.0 Related Best Practices

N/A.

8.0 Acknowledgements

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